



Apple buys Camel Audio

Camel Audio, the developer of Alchemy Software Synth has been purchased by Apple to improve its audio expertise <http://dgit.in/AppCamel>



You see social media sites having these kind of numbers, but we physically have these many people walking in and dealing with us. We have complex problems both online and the physical stores, dealing with our register systems, you know tens of thousands of transactions in a second during black Fridays or holidays. It's a great place to work if you want to work on technical challenges.

How do you work on this concurrent access problem? Here in India we've had a lot of launches recently – products that are really hot – and tens of thousands of people clicking at the same time to grab them – how do you resolve that? It's a major challenge here as well.

Jeremy: I've spent seven years in eBay and that's exactly the problem that eBay had. What's happened over the last 10 years is that we have seen a change in the amount of technology that you're able to use (even opensource technology) that can handle these type of scale problems. You look at the things that have been opensourced by LinkedIn, Facebook, Netflix and Walmart also open sources some of our technologies as well. There's been a rich history of building these things at these massive scales. We have these problems like crazy. Think about an Xbox on a Black Friday, how do you tell an online customer, how many Xbox consoles are there at the store on a black Friday because they're going off the shelf as fast as you can put them on. In addition to that units are all on customer's baskets and this is where some day, the internet of things can tell where in the store every single Xbox is.

From the technology standpoint, when it comes to keeping track of at least high value items how do you currently figure out where they are in a supply chain? Do you already use something like a RFID?

Jeremy: No, not yet, we have started experimenting with in our Sam's Club business, it is a little bit easier because it's a lot more assortment, less items in Sam's Club. You can really think about mapping the whole store real time. Mostly what we're interested in is things like customer behavior, when I come into a Sam's club and I want to pick up my order, often

times it's a palate of things. I'm reordering for my store, I'm reordering for my school, business and office supplies, and I don't want to stand there for 20 minutes while they bring the stuff out. There could be the Uber-like check ins at locations. Imagine you're like "hey I'll be at the store in like 20 minutes have it all ready." So we built kiosks for people who come in, mobile apps that allow you to do that. Same thing in Walmart, when your pick up item is ready, it will give you a notification that it is ready. I think in the long term there's a lot more work that needs to be done in order to do package tracking, on a massive scale. We built a product last year called Scanning that allows the customer to come in and effectively scan items and put them in your carts. When you go to the register, you scan what's on your list, it automatically comes to the register and then you just leave.

Someone in Korea had done that right?

Jeremy: Oh that was a virtual wall and we had that in China in Walmart as well and we've had that in the US but it was essentially a virtual environment where you go beep beep beep scan items off a wall, that sort of thing. In the UK for example, in the London Tube they have these ordering stations that you order on your mobile device in the morning and then when

Analytics (laughs). The mantra behind it is dealing with price and taking costs out of the system. Online this is an interesting problem, because a consumer has so many retailers just a click away, we want to make sure that we have the low price across the internet to remain competitive. How do you react to price changes? We've built not only the ability to understand what price other online retailers are selling things for but we also built a system called smart pricing that effectively takes that data plus our cost data, our score data, and an algorithm that effectively allows items to be automatically priced. To make sure we are not losing out.

Let's look into the future and know according to you, what should be the buying experience should be 5-10 years from now? What's the ideal experience?

Jeremy: I've seen some prototypes of devices with which you can take a picture of your cabinet and it'll know you're out of things. Also most people shop from a list and Walmart is smart enough to know that you shop with us enough, we know the frequency of when you need items. Shopping could soon be a really personalized experience where you get suggestions about how people like you also bought this, or we'll tell you we have a new special offer on a particular item, or the fact that other people

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you come back in the night, you have your two bags of items waiting for you. You just pick them up and walk in. It has really been great. The nice part of having a global footprint is you can experiment all over the world and take this capability into markets that make sense.

Any interesting thing in big data analytics that you guys are doing?

Jeremy: Yes, the Bangalore team in particular over the last couple of years has built out a capability we call the CIA, stands for Competitive Intelligence and

in your area have liked something. We can also see that when you shop online for your groceries, you have 80% of the stuff being bought again and again those will be automatically added in for you. We have already done a lot of this at Sam's Club and in Walmart on our website. You can see a lot more of that coming in the future. **1**

To read the full review where we talk about the future of shopping, the "Target Incident", and more, follow this link: dgit.in/jrmykng1 or Blippar this page.